CO-PO Attainment Handbook

By

Internal Quality Assurance Cell (IQAC)

CO-PO Attainment Handbook

Terminology (Abbreviations)

- **OBE:** Outcome-Based Education (OBE) is a student-centric teaching and learning methodology in which the course delivery, assessment are planned to achieve stated objectives and outcomes. It focuses on measuring student performance i.e. outcomes at different levels.

- **Course Outcomes (CO)**: Course Outcomes (COs) are what the student should be able to do at the end of a course. The most important aspect of a CO is that it should be observable and measurable

- **Program Outcomes (PO)**: Program outcomes are statements that describe what the knowledge, skills and attitudes students should have at the time of graduation from an engineering program. That means just at the end of 4 years these represent what is the knowledge, skills and attitudes they should have. And at present POs are 12 in number and they are identified by NBA and are applicable to all engineering programs.

- **Program Educational Objectives (PEO**) : These are broad statements that describe the career and professional accomplishments in four to five years after graduation that the program is preparing the graduates to achieve

- **Program Specific Outcomes (PSO):** PSOs are outcomes that are specific to a program. PSOs characterize the specificity of the core courses of a program. PSOs can be 2 to 4 in number.

Few definitions

Mapping Factor (Correlation Level)

It indicates to what extent a certain component (either assessment method to CO or CO to PO or PO to PEO & PSO

- **3-indicates Substantial (high)** mapping (high contribution towards attainment)
- **2-indicates Moderate** (medium) mapping (medium contribution towards attainment)
- **1-indicates Slight (low)** mapping (some contribution towards attainment)

Level of attainment

Here 3 levels of attainment is taken as 1-Low; 2-medium; 3- High

3 levels of attainment can be defined as

- Attainment 3 : 60% Stud scoring >= 70% of max marks allocated to CO
- Attainment 2 : 50% Stud scoring >= 70% of max marks allocated to CO
- Attainment 1 : 40% Stud scoring >= 70% of max marks allocated to CO
- Attainment 0 : Less than 40% Stud scoring >=70% of max marks allocated to CO

Setting CO Attainment Targets

There can be several methods

Example 1:

- Same target is identified for all the COs of a course.
- For example the target can be "the class average marks ≥ 60 marks"

Example 2

- Targets are the same for all COs and are set in terms of performance levels of different groups of students.
- While this method classifies students into different categories, it does not provide any specific clues to plans for improvement of quality of learning

Targets			
(% of students getting < 50)	(% of students getting >50 and < 65)	(% of students getting >65 and < 80)	(% of students getting ≥ 80)
10	40	40	10

Example 3:

- Targets are set for each CO of a course separately
- It does not directly indicate the distribution of performance among the students. However, it has the advantage of finding out the difficulty of specific COs

CO	Target (Class Average)
CO1	70%
CO2	80%
CO3	75%
CO4	65%
CO5	70%
CO6	80%

Example 4:

- Targets are quantized in to certain levels, 3 being the most common number of levels.
- Level 3: 60% Stud scoring >= 70% of max marks allocated to CO
- Level 2 : 50% Stud scoring >= 70% of max marks allocated to CO
- Level 1 : 40% Stud scoring >= 70% of max marks allocated to CO
- Level 0 : Less than 40% Stud scoring >=70% of max marks allocated to CO
- Aim is to attain Level 3

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Attainment of COs

- Attainment of COs can be measured **directly** and **indirectly**
- Direct attainment of COs can be determined from the performances of students in all the relevant assessment instruments.
- Indirect attainment of COs (which is optional as per NBA) can be determined from the course exit survey.
- The exit survey form should permit receiving feedback from students on all the COs.

Direct CO attainment

- Direct attainment of COs is determined from the performances of students in Continuous Internal Evaluation (CIE) and Semester End Examination (SEE).
- The proportional weightages of CIE: SEE will be as per the academic regulations in force. Proportions of 20:80, 40:60 are all possible!
- Direct attainment of a specific COs is determined from the performances of students to all the assessment items related to that particular CO.
- Hence, every assessment item needs to be tagged with the relevant CO.
- Also, we need data about performance of students assessment item – wise.

Direct CO attainment from CIE

- Continuous Internal Evaluation (CIE) is conducted and evaluated by the Department itself Thus, institution have access to question-wise marks in all assessment instruments in CIE.
- When questions are tagged with relevant COs, the department has access to performances of students with respect to each CO.
- Hence, computing the direct attainment of COs from CIE is straight forward for Tier 2 institutes.

Direct CO attainment from SEE

- However, Semester End Examination (SEE) is conducted and evaluated by the University for Tier 2 institutes.
- Thus the departments in Tier 2 institutes get only total marks scored in SEE and not question-wise marks!
- As a consequence, departments in Tier 2 institutes have no means of computing the direct attainment of individual COs from SEE!
- SEE performance cannot be ignored!!
- The only possible solution, though not satisfactory, is to treat the average marks in SEE as the common attainment of all COs!!!

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CO attainment Computation:

STEP-1 : For every subject 4-7 course outcomes (CO) are defined and mapped to Program outcomes (PO) on a scale of 0 to 3. Highest correlation is 3. For example,

CO's to PO's and PSO's Mapping

Subject	SMS																
Code	15CS834																
CO	COURSE OUTCOMES	Modules Covered	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	Explain the system concept and apply functional modeling method to model the activities of a static system	1,2	2	3	1	1	-	-	-	-	-	-	-	-	-	2	-	1
CO2	Describe the behavior of a dynamic system and create an analogous model for a dynamic system	1,2	2	3	1	1	-	-	-	-	-	-	-	-	-	2	-	1
CO3	Simulate the operation of a dynamic system and make improvement according to the simulation results	1,2,3,4,5	2	3	1	1	-	-	-	-	-	-	-	-	-	2	-	1

STEP 2

Maximum marks allotted to each question, mapped to a cognitive level and the corresponding CO. Record the percentage of students achieving a set percentage of max marks allotted to an individual CO in a given IAT.

For example,

	Revised Bloom's Levels (L1,L2,L3,L4,L5,L6)		L2	L2		L2			L2	L2		L2	L2	
	Question maps to which course outcome?		CO1	CO1		CO1			CO2	CO2		CO2	CO2	
Subject:	Data Structures and Applications		IAT-1											
Code	15CS33	MAX MARKS	5	5		10			5	5		5	5	
		Attainment	3	0	NA	1	NA	NA	3	0	NA	3	3	NA
		Attempted Count	36	28	0	31	0	0	9	6	0	40	37	0
		Student Scored More than 70%	25	8	0	15	0	0	6	2	0	35	32	0
Sl. No.	Student USN	Name	Q1A	Q1B	Q1C	Q2A	Q2B	Q2C	Q3A	Q3B	Q3C	Q4A	Q4B	Q4C
1	1CR17CS136	SHIVAM KUMAR		3		4						5	5	
2	1CR17CS138	SHIVAPRASAD S B		8								5	5	
3	1CR17CS140	SHREYA C BHATT	5	2		5						2	5	
4	1CR17CS141	SHREYA R	5	2		8			3	1		4	5	
5	1CR17CS142	SHREYAS KILINGARU NA	1	3		10						5	5	
6	1CR17CS143	SHRUTI SINGH D/O Nagendra Singh	0			8						5	5	
7	1CR17CS144	SINGH	2	4		9			3	1		5	5	
8	1CR17CS145	SHUBHAM GOYAL	5	3		10						5	4	

	Revised Bloom's Levels (L1,L2,L3,L4,L5,L6)		L3			L1	L2		L1	L2		L3	L3		
	Question maps to which course outcome?		CO2			CO2	CO1		CO1	CO1		CO2	CO2		
Subject:	Data Structures and Applications														TOTAL
Code	15CS33	MAX MARKS	10			5	5		4	6		5	5		80
		Attainment	0	NA	NA	3	0	NA	3	2	NA	2	2	NA	0
		Attempted Count	11	0	0	21	9	0	46	44	0	39	23	0	48
		Student Scored More than 70%	4	0	0	15	2	0	43	25	0	23	13	0	0
Sl. No.	Student USN	Name	Q5A	Q5B	Q5C	Q6A	Q6B	Q6C	Q7A	Q7B	Q7C	Q8A	Q8B	Q8C	
1	1CR17CS136	SHIVAM KUMAR							2	3		4	4		30
2	1CR17CS138	SHIVAPRASAD S B	10						5	5		5	5		48
3	1CR17CS140	SHREYA C BHATT							5	4		3	2		33
4	1CR17CS141	SHREYA R	1			5			5	4		1	1		45
5	1CR17CS142	SHREYAS KILINGARU NA							5	4		4	3		40
6	1CR17CS143	SHRUTI SINGH D/O Nagendra Singh				4			5	5		5	5		42
7	1CR17CS144	SINGH				5	2		5	5		5	4		55
8	1CR17CS145	SHUBHAM GOYAL	9						4	5					45

STEP 3

Two best performances of a student from three IATs are used for calculating attainment levels for CO1. The process is described below.

Condition	
IF S3 % of students score ≥ M3% of Max marks allotted to CO - Att. Lev. 3	
ELSE IF S2% of students score ≥ M2% of Max marks allotted to CO - Att. Lev. 2	
ELSE IF S1% of students score ≥ M1% of Max marks allotted to CO - Att. Lev. 1	
ELSE Att. Lev. 0	

In our case we have taken S3,S2,S1 as 60%,50%,40% and M3,M2,M1 as 70%

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STEP 4

Repeat the above condition to evaluate all COs.

Revised Bloom's Levels (L1,L2,L3,L4,L5,L6)		L3	L2	L2	L3	L3	L3	L3	L2	L3	L2	L2	L3	L3	L1		
		CO1	CO1	CO1	CO1	CO1	CO1	CO1	CO1	CO1	CO1	CO2	CO2	CO2	CO2		
IAT-1 2017-18 EVEN SEM																	
	MAX MARKS	7	3	6	4	7	3	4	6	7	3	6	4	7	3		
USN	Name	Q1A	Q1B	Q2A	Q2B	Q3A	Q3B	Q4A	Q4B	Q5A	Q5B	Q6A	Q6B	Q7A	Q7B		
C-02	AADITYA RANJAN	7	3	6	2	4		1.5	4			5	1	2			
C-03	AAYUSH LAL ROY	5		6	2	6		4	6	7	1.5				1		
C-11	ARCHITH VINOD	5	3	6	4	7		3	3			5	1				
#Stud >0		5	3	5	4	5	1	5	5	1	1	2	2	2	2		0
70% of Max Marks		4.9	2.1	4.2	2.8	4.9	2.1	2.8	4.2	4.9	2.1	4.2	2.8	4.9	2.1	0	0
# Stud > Max Marks		4	2	4	1	3	1	2	3	1	0	2	0	1	0	0	0
%Stud > Max Marks		80	66.7	80	25	60	100	40	60	100	0	100	0	50	0		
Attainment Level		3	3	3	0	2	3	0	2	3	0	3	0	1	0	0	0
CO1 Attainment		2.11															
CO2 Attainment		1															

STEP 5

Calculate the attainment levels based on internal test and VTU University Examinations using the below formula.

For External Exam	**For Internal Exam**
60% Stud >= 55% of max marks : ATT 3	60% Stud >= 55% of max marks : ATT 3
50% Stud >= 55% of max marks: ATT 2	50% Stud >= 55% of max marks: ATT 2
40% Stud >= 70% of max marks : ATT 1	40% Stud >= 70% of max marks : ATT 1
else ATT 0	else ATT 0

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Externals

Subject: Code	15CS834	SMS MAX MARKS	External 80
		Attainment	3
		Attempted Count	103
		Student Scored More than 55%	77
SN	USN	Name	Marks
1	1CR14IS120	PRIYANKA V	34
2	1CR14IS026	AYUSH GOLYAN	52
3	1CR15IS006	ADITYA CHOUDHARY	45
4	1CR15IS020	DARSHAN M JAIN	57
5	1CR15IS082	RIYA REGI	52
6	1CR15IS012	ASHWINI B L	45
7	1CR15IS109	THANKAM PAUL	59
8	1CR15IS017	CHARUMATHI N	34
9	1CR15IS078	RASHI SRIVASTAVA	53
10	1CR15IS068	PRAKHAR TARUN	52
11	1CR15IS060	NAGA SANITH REDDY P	28
12	1CR15IS054	MANIKANTA G	45
13	1CR15IS050	LAKSHMI GIRIJA MANGIPUDI	46
14	1CR15IS035	ISHA AGARWAL	40

………………………..

AQSM

Revised Bloom's Levels (L1,L2,L3,L4,L5,L6)			
Question maps to which course outcome?			CO1,CO2,CO3
Subject: Code	15CS834	SMS MAX MARKS	AQSM 5
		Attainment	3
		Attempted Count	103
		Student Scored More than 70%	103
SN	USN	Name	Final Assignment Marks
1	1CR14IS026	Ayush golyan	5
2	1CR14IS033	Deepika S	5
3	1CR14IS120	Priyanka V	5
4	1CR15IS001	ABHISHEK KUMAR	5
5	1CR15IS002	ABHISHEK KUMAR	5
6	1CR15IS003	Abhishek Raj	5
7	1CR15IS005	Aditi Sharma	5
8	1CR15IS006	Aditya Choudhary	5
9	1CR15IS007	Aishwarya M	5
10	1CR15IS008	Akshat Kumar Singh	5

<u>STEP 6</u>

CO attainment level for the that course is,

Course attainment @Internals = 0.75 * Avg IAT attainment + 0.25 * AQSM

<u>For 2015 and 2016 Scheme :</u>

Final $\boldsymbol{Course\ attainment\ level\ (CAL) = (0.8 \times External\ att.) + (0.2 \times Internal\ att.)}$

<u>For 2017 Scheme and 2018 Scheme:</u>

COURSE ATTAINMENT LEVEL calculation = (0.6 * external attainment) + (0.4 * Int. attainment)

CO's Attainment

SubJect:	SMS							
Code:	15CS834							
CO's	IAT-1 Attainment	IAT-2 Attainment	IAT-3 Attainment	AVG IAT Attainment	AQSM Attainment	INTERNAL Attainment	External Attainment	Final Attainment(80%
C01 Attainment	2	NA	3	2.5	3	**2.6**	3	**2.9**
C02 Attainment	3	3	NA	3	3	**3.0**	3	**3.0**
C03 Attainment	NA	2.7	2.5	2.6	3	**2.7**	3	**2.9**
Average Attainment								**3.0**

STEP 7

Program outcomes attained through the attainment of COs. For a given course, all COs are mapped to certain POs, as shown in STEP 1. The overall CO attainment value as computed in STEP 7 and the CO-PO mapping values given in the STEP 1 used to compute the attainment of POs.

CO-PO attainment:

PO attainment can be computed for a batch using the below formula.

PO/PSO attainment = (CO attainment * CO-PO Mapping)/Max correlation strength

CO-PO and PSO Attainments

Subject:	SMS															
Code	15CS834															
Course Name	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
C01	1.95	2.9	0.975	0.975	-	-	-	-	-	-	-	-	-	1.95	-	0.975
C02	2	3	1	1	-	-	-	-	-	-	-	-	-	2	-	1
C03	1.96	2.9	0.98	0.98	-	-	-	-	-	-	-	-	-	1.96	-	0.98
Final Attainment	**1.97**	**3.0**	**0.98**	**0.98**										**1.97**		**0.98**

i.e. CO1 attainment final =2.9 and CO1-PO1 mapping is 2, so the attainment w.r.t CO1 and PO1 is = (2.9 * 2)/3 = 1.93

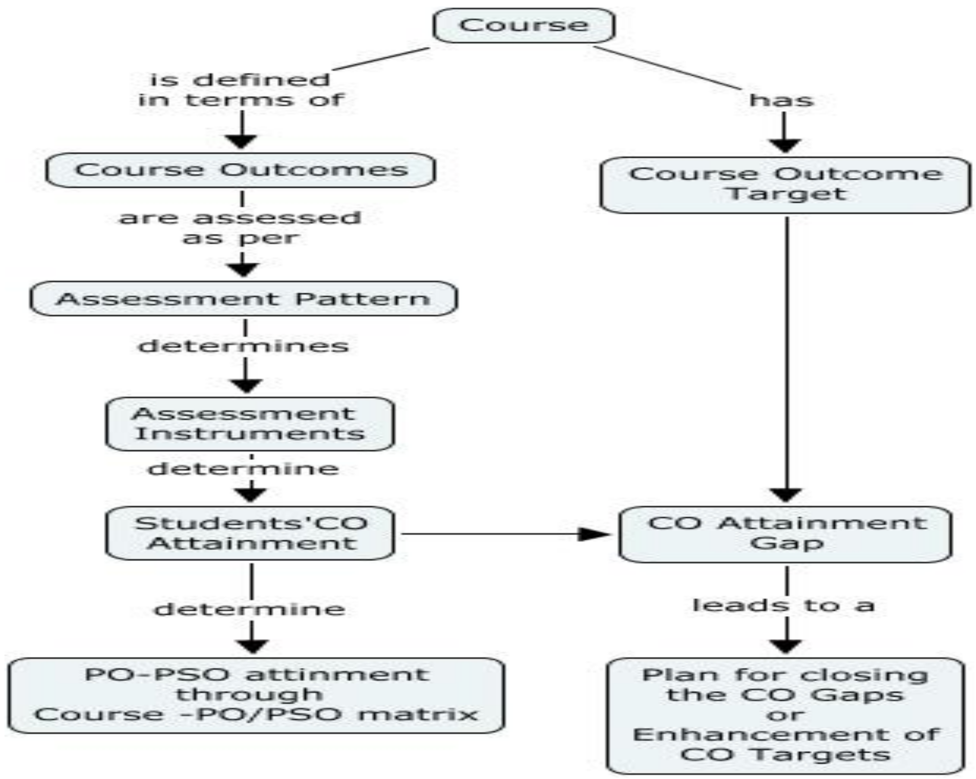
CO1-PO2 mapping is 3 hence CO1-PO2 = (2.9*3)/3 = 2.9

Same process is repeated for all the POs.

Final CO attainment w.r.t PO and PSO

Final CO attainment value	CO-PO and CO-PSO attainment															
	CO-PO1	CO-PO2	CO-PO3	CO-PO4	CO-PO5	CO-PO6	CO-PO7	CO-PO8	CO-PO9	CO-PO10	CO-PO11	CO-PO12	CO-PSO1	CO-PSO2	CO-PSO3	CO-PSO4
3	1.97	3	0.98	0.98	-	-	-	-	-	-	-	-	-	1.97	-	0.98

CO attainment and Gap Analysis



Couse Outcome	Attainment Level for last year	Target for current exam	Attainment Level of current year	Gap	Gap Analysis
CO1	2.8	2.9	2.9	-	No gap, all students have cleared in externals
CO2		2.9	3	-	
CO3		2.9	2.9	-	

STEP 8

PO attainment can be computed for a batch using the below formula. Indirect attainment is determined from student exit surveys, employer surveys, co-curricular activities, extracurricular activities and mapped to POs. A questionnaire was designed for this purpose and the average response of the outgoing students for each PO is computed.

Final PO attainment for a particular batch = 0.8 * Direct Attainment + 0.2 * Indirect attainment

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		PO Attainment													PSO Attainmer		
SN	Subject	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2	PSO3
1	Engineering Mathematics-I	0.89	1.33	-	-	-	-	-	-	-	-	-	-		0.89	1.33	-
2	Engineering Physics	0.93	0.47	-	0.47	-	-	0.93	0.93	0.93	-	-	-		0.93	0.47	-
3	Elements Of Civil Engineering And Mechanics	1.59	0.73	-	1.47	1.47	0.73	-	0.73	0.73	0.73	1.47	2.20		1.59	0.73	-
4	Elements Of Mechanical Engineering	0.40	0.40	0.20	-	-	-	-	-	-	-	-	-		0.40	0.40	0.20
5	Basic Electrical Engineering	0.36	0.32	-	-	-	-	-	-	-	-	-	-		0.36	0.32	-
6	Workshop Practice	0.43	0.43	-	-	0.40	0.20	0.20	0.24	0.20	0.33	0.40	0.50		0.43	0.43	-
7	Engineering Physics Lab	0.43	0.43	0.37	0.43	0.40	0.20	0.20	0.24	0.20	0.33	0.40	0.50		0.43	0.43	0.37
8	Constituttion Of India, Professional Ethics And Human Rights	-	-	-	-	-	-	-	2.80	2.80	2.80	1.87	2.80		-	-	-
9	Engineering Mathematics – II	2.20	0.73	-	0.73	-	-	-	-	-	-	-	0.73		2.20	0.73	-
10	Engineering Chemistry	0.93	0.47	0.47	-	-	-	-	-	-	-	-	-		0.93	0.47	0.47
11	Programming In C And Data Structures	0.47	0.47	-	0.47	-	0.47	-	-	-	-	-	-		0.47	0.47	-
12	Computer Aided Engineering Drawing	2.00	1.00	1.50	1.00	2.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00		2.00	1.00	2.00
13	Basic Electronics	0.40	-	-	-	0.20	-	-	-	0.20	-	-	-		0.37	0.40	-
14	Computer Programming Language Lab	1.00	1.00	-	-	-	1.00	-	1.00	-	-	-	-		1.00	1.00	-
15	Engineering Chemistry Lab	0.43	0.43	0.37	0.43	0.40	0.20	0.20	0.24	0.20	0.33	0.40	0.50		0.43	0.43	0.37
16	Environmental Studies	-	-	-	-	-	-	-	3.00	3.00	3.00	2.00	3.00		-	-	-
17	Engineering Mathematics - III	-	1.47	-	1.47	-	-	-	-	0.73	-	-	1.47		0.73	0.73	0.73
18	Analog and Digital Electronics	0.20	0.20	0.16	-	-	-	-	-	-	-	-	-		-	-	0.13
19	Data Structures and Applications	0.50	0.40	0.40	0.60	0.20	0.40	0.40	0.20	0.40	0.40	0.40	0.50		0.40	0.40	0.60
20	Computer Organization	0.40	0.27	-	0.13	0.13	-	-	0.13	0.13	0.27	0.13	0.22		0.13	0.13	0.27
21	Unix and Shell Programming	0.67	-	-	0.67	0.67	-	-	0.67	0.67	0.67	0.67	1.33		0.67	-	1.33
22	Discrete Mathematical structures	2.20	1.47	0.98	1.47	0.73	0.73	-	-	0.73	-	-	-		-	-	0.73
23	Analog and Digital Electronics Laboratory	1.50	2.50	3.00	-	-	-	-	-	-	-	-	1.00		1.00	2.00	-
24	Data Structures Laboratory	2.00	2.00	2.00	2.00	1.00	1.00	1.00	-	-	1.00	1.00	1.00		3.00	2.00	2.00
25	Engineering Mathematics - IV	1.89	2.00	0.67	-	-	-	-	-	0.67	-	-	-		0.67	0.67	0.67
26	Software Engineering	1.40	0.93	0.47	0.47	0.47	-	-	-	-	-	0.47	-		0.65	0.47	0.84
27	Design & Analysis of Algorithms	0.86	1.34	1.47	1.34	1.22	0.73	-	-	1.47	0.73	0.86	0.86		0.98	1.47	1.32
28	Microprocessor and Microcontroller	0.60	0.20	-	-	-	-	-	-	-	0.20	-	-		0.35	0.33	0.20
29	OBJECT ORIENTED CONCEPTS	0.60	0.56	0.60	0.56	-	-	-	-	0.40	-	-	0.56		0.32	-	0.20
30	Data Communication	0.20	0.40	0.60	-	-	-	-	-	-	0.20	-	-		0.33	0.44	0.20
31	Design & Analysis of Algorithms Lab	2.20	2.20	2.20	1.47	0.73	1.47	-	-	0.73	0.73	-	1.47		0.73	0.73	-
32	MICROPROCESSOR AND MICROCONTROLLER. LABORATORY	0.67	0.33	1.27	0.33	0.67	-	-	-	0.33	0.33	-	0.33		0.33	-	1.00
33	MANAGEMENT AND ENTREPRENEURSHIP FOR IT INDUSTRY	1.00	2.00	1.67	1.00	1.00	1.17	1.00	1.00	2.00	1.17	1.50	2.00		2.00	-	1.00
34	COMPUTER NETWORKS	1.47	1.47	1.96	-	0.73	-	-	0.73	0.73	0.73	-	1.47		-	0.73	0.86
35	DATABASE MANAGEMENT SYSTEM	1.47	1.47	1.10	1.22	1.47	0.73	-	-	1.10	0.86	0.73	1.47		1.47	-	-
36	AUTOMATA THEORY AND COMPUTABILITY	1.47	1.61	1.17	1.03	0.73	0.73	-	-	0.73	0.98	-	0.73		0.73	0.73	0.73
37	Advanced Java & J2EE	2.00	1.00	2.00	-	1.00	-	-	-	1.00	1.00	-	2.00		1.97	-	2.00
38	AUTOMOTIVE ELECTRONICS	2.80	1.63	-	-	-	-	-	-	-	-	-	-		-	-	-
39	8051 MICROCONTROLLER	0.80	0.53	0.53	0.27	-	-	-	-	-	0.27	-	-		-	-	-
40	OPTIMIZATION TECHNIQUES	2.18	2.18	1.87	2.18	-	-	-	-	-	-	-	-		-	-	3.00
41	TRAFFIC ENGINEERING	1.49	0.93	0.93	0.93	0.93	0.93	1.87	-	-	0.93	0.93	0.93		-	0.93	-
42	AUTOMATION AND ROBOTICS	2.00	2.33	2.00	-	2.00	-	-	-	-	-	-	-		2.00	-	-
43	COMPUTER NETWORK LAB	0.70	2.00	3.00	1.00	3.00	-	2.00	-	-	-	-	1.00		-	1.20	0.80
44	DBMS LABORATORY WITH MIN PROJECT	1.00	2.00	-	2.00	2.00	3.00	1.00	-	-	-	-	-		1.00	-	-
45	CRYPTOGRAPHY, NETWORK SECURITY AND CYBER LAW	0.93	0.86	0.70	0.93	0.47	0.93	0.47	0.70	0.54	0.47	0.47	0.93		0.47	0.47	0.93
46	FILE STRUCTURES	0.48	0.56	0.60	0.20	0.40	-	-	-	0.40	-	-	-		0.40	-	0.40
47	SOFTWARE TESTING	0.73	1.32	1.10	0.88	0.73	-	-	-	0.73	0.73	0.73	1.17		-	-	2.20
48	OPERATING SYSTEMS	2.80	1.00	-	1.00	1.00	-	-	-	1.00	-	-	1.00		-	-	1.00
49	DATA MINING AND DATA WAREHOUSING	1.32	1.76	0.98	0.98	1.47	0.92	1.10	0.73	1.10	-	-	1.03		0.73	-	0.73
50	OPERATION RESEARCH	2.00	2.00	2.00	1.00	1.00	1.00	1.00	-	-	-	-	-		2.00	-	1.00
51	MOBILE APPLICATION DEVELOPMENT	0.40	0.20	0.28	0.24	0.27	0.40	-	0.40	0.20	0.20	-	0.20		0.27	-	-
52	BIG DATA ANALYTICS	0.40	0.30	0.50	0.35	0.30	0.45	0.40	0.40	0.40	-	-	-		0.50	-	0.20
53	PYTHON APPLICATION PROGRAMMING	2.83	2.33	2.17	1.50	2.00	1.00	-	-	2.00	1.00	1.00	2.00		2.00	2.00	1.00
54	SOFTWARE TESTING LABORATORY	1.00	3.00	-	1.00	1.00	1.00	-	-	2.00	1.00	-	-		3.00	2.00	2.00
55	File Structures Lab	1.00	2.00	1.00	2.00	-	1.00	-	-	-	-	-	-		2.00	2.00	2.00
56	Web Programming & its Applications	0.73	0.73	1.47	0.73	0.88	1.47	-	-	1.47	0.73	1.47	0.88		2.20	-	-
57	Software Architecture & Design Patterns	1.10	1.83	1.71	-	-	0.73	-	-	-	-	-	-		-	-	1.47
58	MACHINE LEARNING	0.93	0.93	0.93	0.93	0.93	0.65	0.65	0.93	1.40	1.40	-	0.93		1.47	-	1.47
59	Natural Language Processing	2.20	1.47	1.28	-	-	1.47	-	-	-	-	-	-		1.47		
60	Cloud Computing	1.71	0.86	0.86	0.90	1.58	1.44								1.71		
61	STORAGE AREA NETWORKS																
62	WEB TECHNOLOGY LABORATORY WITH MINI Project																
63	Machine Learning Lab																
64	Internet of Things Technology																
65	Big Data Analytics	2.70	2.90	2.90	2.70	2.70	2.70								2.80		2.70
66	System Modeling and Simulation	2.80	3.00	2.60	2.60											2.77	
67	Project Work	2.80															
68	Seminar	2.80															
69	Internship																
	Average	1.29	1.22	1.26	1.03	1.01	0.96	0.84	0.85	0.92	0.85	0.89	1.14		1.10	0.91	1.03

	PO Attainment													PSO Attainmer		
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**		**PSO1**	**PSO2**	**PSO3**
ISE(Direct)	**1.29**	**1.22**	**1.26**	**1.03**	**1.01**	**0.96**	**0.84**	**0.85**	**0.92**	**0.85**	**0.89**	**1.14**		**1.10**	**0.91**	**1.03**
ISE (Indirect)	**2.02**	**1.95**	**2.2**	**2.05**	**2.07**	**2.29**	**2.26**	**2.21**	**2.14**	**2.05**	**2.17**			2.20	2.07	2.14
Final attainent	**1.43**	**1.37**	**1.44**	**1.23**	**1.22**	**1.23**	**1.12**	**1.12**	**1.17**	**1.09**	**1.15**	**0.91**	**0.00**	**1.32**	**1.14**	**1.25**

In-Direct Survey

Questionnaire for the Assessment

At the end of the program I am able to

Kindly give your responses on the following outcomes.

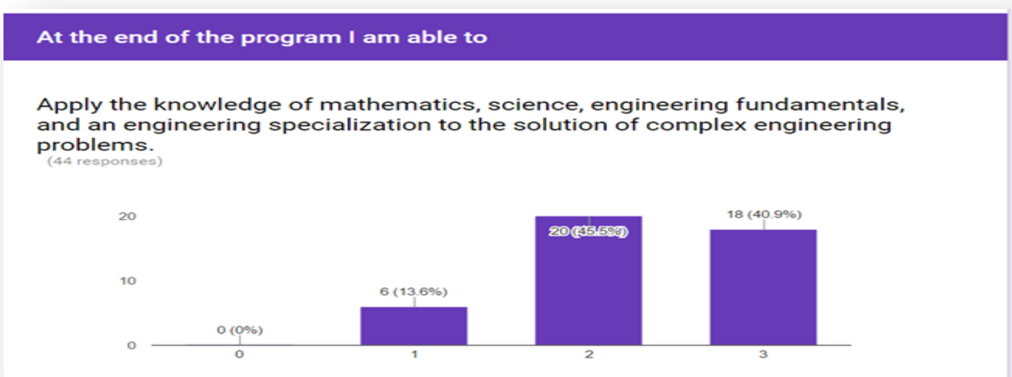
Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

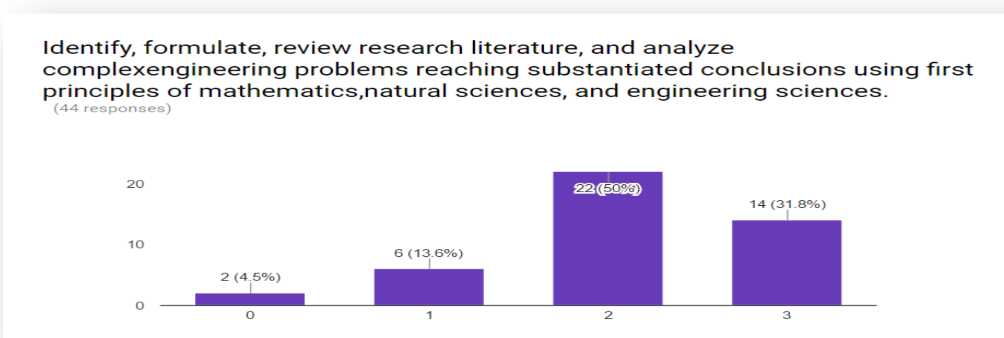
	0	1	2	3	
Don't agree	○	○	○	○	Strongly agree

Identify, formulate, review research literature, and analyze complexengineering problems reaching substantiated conclusions using first principles of mathematics,natural sciences, and engineering sciences.

	0	1	2	3	
Don't agree	○	○	○	○	Strongly agree

Responses





PO and PSO attainment Batch Wise –Analysis

PO and PSO attainment analysis: 2015-19 batch

POs	Target Level	Attainment Level	Observation
PO1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.			
PO1	**1.82**	**1.53**	Requires awareness of Mathematics and Engineering fundamentals in Engineering problems.
Action 1: Extra classes to be conducted for slow learners beyond the regular planned classes. **Action 2:** Additional Maths classes are conducted during the semester after every internal based on the performance. **Action 3:** Additional topic specific tests have been conducted.			
PO2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems in Civil Engineering reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.			
PO2	**1.72**	**1.52**	The scheme changed from 2010 scheme to 2015 scheme with this batch hence targeted attainment fell short maybe due to newer curriculum introduced for 1st time.
Action 1: Hackathon events are conducted, where the students are exposed to latest technologies. **Action 2:** Question bank comprising of important questions from typical previous year question papers were prepared and attached to department repository which was accessible to students for further self-improvement. Action 3: During lab students complete the extra programs other thanVTU syllabus which helps them to increase analytical skills			
PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes of Civil Engineering that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.			
PO3	**1.73**	**1.53**	
Action 1: Students are motivated to develop mini-projects focusing on real world problems.			

Action 2 : Students are encouraged to participate in various coding competitions which involves the design and development of solutions. Action 3 : Hackathon events are conducted, where the students are exposed to latest technologies and develop solutions for the real world problems.			
PO4: Conduct investigations of complex problems: Use research based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.			
PO4	**1.68**	**1.27**	Attainments are required to be improved by increasing student ability to analyze the complexity of the problem
Action 1: More projects on solving complex problems will be focused. Action 2: Mini projects given to students are quite complex to help them in understanding complex problems.			
PO5 : Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.			
PO5	**1.63**	**1.38**	Needs to enhance students' knowledge to create, select, and apply in appropriate techniques using IT tools.
Action 1 : Students are exposed to different open source software during their lab sessions. Action 2: Made the students to do projects using modern tools like Android programming, Internet of things. Action 3 : Additional guest lectures and workshops are conducted to educate students on modern IT tools.			
PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.			
PO6	**1.31**	**1.26**	Interaction with the society need to be improved.
Action 1 : Students are motivated to be part of CSI chapter, RISE club and make them actively participate and be members of such groups.			
PO7: Environment and sustainability: understand the impact of the professional engineering solutions in societal and environmental contexts, demonstrate the knowledge of, and need for sustainable development.			
PO7	**1.32**	**1.22**	

<table>
<tr><td colspan="4">Action 1: Encourages students to take part in competitions,major projects,industry visits, relevant guest lecture.</td></tr>
<tr><td colspan="4">PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.</td></tr>
<tr><td>PO8</td><td>1.54</td><td>1.23</td><td></td></tr>
<tr><td colspan="4">Action 1 : Ethical hacking workshops are conducted, and students are educated with ethical practices.</td></tr>
<tr><td colspan="4">PO9: Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.</td></tr>
<tr><td>PO9</td><td>1.54</td><td>1.29</td><td></td></tr>
<tr><td colspan="4">Action 1 : Mini-Projects are given to pre- final year students to make them understand the importance of team work and upgrade themselves with latest technologies and in turn to have future industrial exposure.

Action 2 : Students will be encouraged to be a part of various clubs at CMR and take part in the activities to enable them to work in teams.

Action 3 : Final year students are grouped into different teams and are motivated to come up

with innovative projects to solve real world problems.

Action 4 : Encourage students to participate in NSS and other social activities.</td></tr>
<tr><td colspan="4">PO10 : Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions</td></tr>
<tr><td>PO10</td><td>1.55</td><td>1.28</td><td></td></tr>
<tr><td colspan="4">Action 1: TYL classes were conducted on communication skills in first year and assessment will be done after that.

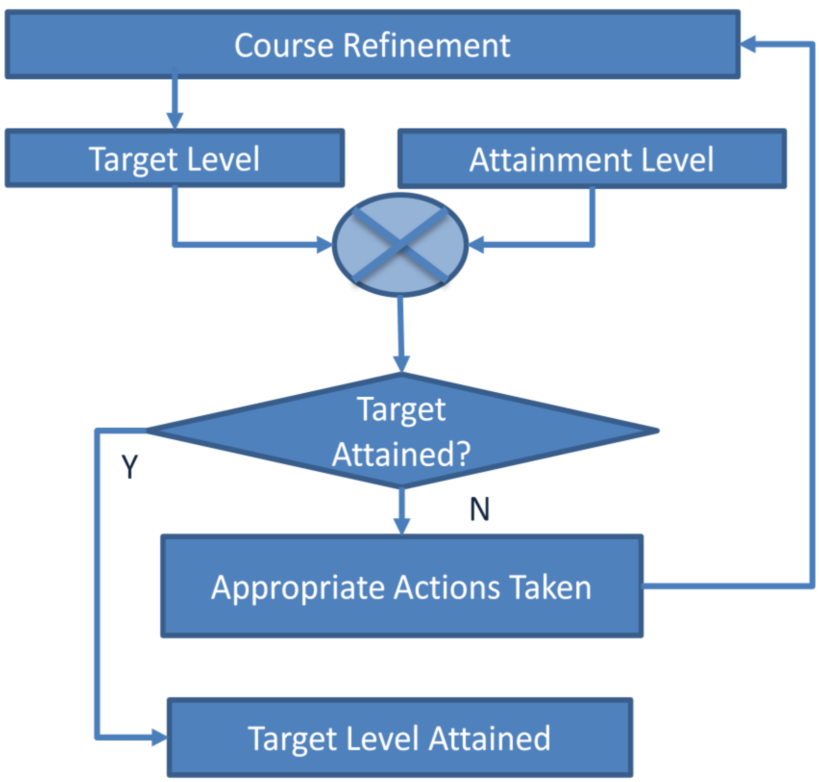
Action 2: Subject Seminars are conducted in the classroom which helps student to improve the communication skills. Even the inclusion of technical seminar will also improve presentation skills.

Action 3: Students are encouraged to write articles for college magazines or papers for technical fests to enhance their documentation skills.

Action 4 : Every subject assignments are mandatory and students are encouraged</td></tr>
<tr><td colspan="4">PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.</td></tr>
<tr><td>PO11</td><td>1.6</td><td>1.27</td><td></td></tr>
<tr><td colspan="4">ACTION 1: Mini projects and projects are done by pre- final and final year students, which</td></tr>
</table>

makes them understand Software Development Life Cycle involved to complete the project. ACTION 2: Time management of students will be judged and assessed through successive completion of projects			
PO12: Lifelong learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.			
PO12	**1.42**	**1.43**	**Target achieved**
Action1: Various Workshops /seminar /guest lectures conducted to educate students on latest technologies. **Action 2:** Every semester students are encouraged to enroll in MOOC Course which motivates them to do lifelong learning.			

Continuous Improvement in PO&PSO Attainment



Every Faculty needs to compute two main attainment values as mentioned below. Based on that if target is not attained hen appropriate actions should be taken.

- Course attainment
- Course w.r.t PO attainment

Department HOD needs to compute batch wise PO and PSO attainment and needs to analyze the gaps and take necessary actions.

IQAC CO-PO Certification Test

202 responses

Publish analytics

Name

202 responses

swathi

Naresh Dixit P S

VINAY M N

GURUPRASAD H C

SRINIVAS REDDY MUNGARA

SHREEKANTH M PRABHU

Dr. Balu P Patil

Imtiyaz Ahmed B K

DINESH R

Department

202 responses

ECE

EEE

CSE

ISE

Mathematics

MBA

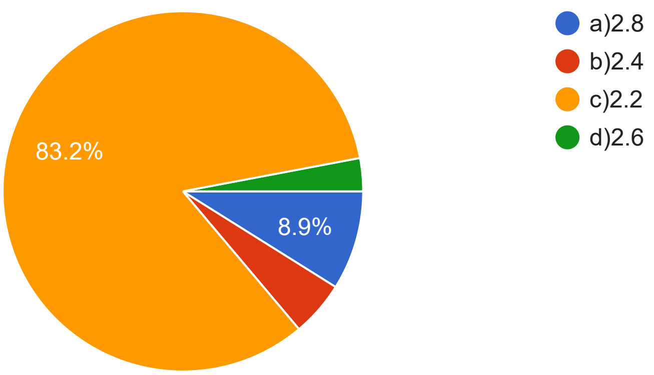
Mechanical Engineering

Physics

Mechanical

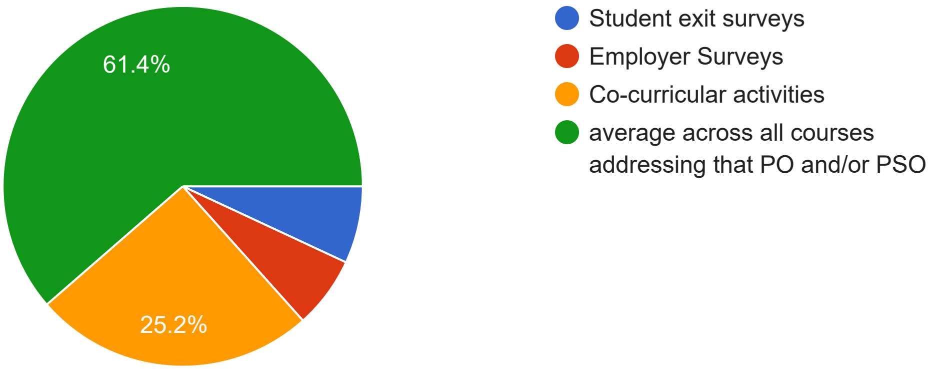
Attainment through University Exam is moderate i.e. 2 and attainment through Internal Assessment is substantial i.e. 3. Assuming 80% weightage to the university exam & 20% weightage to internal assessment, the attained course outcome will be ___.

202 responses



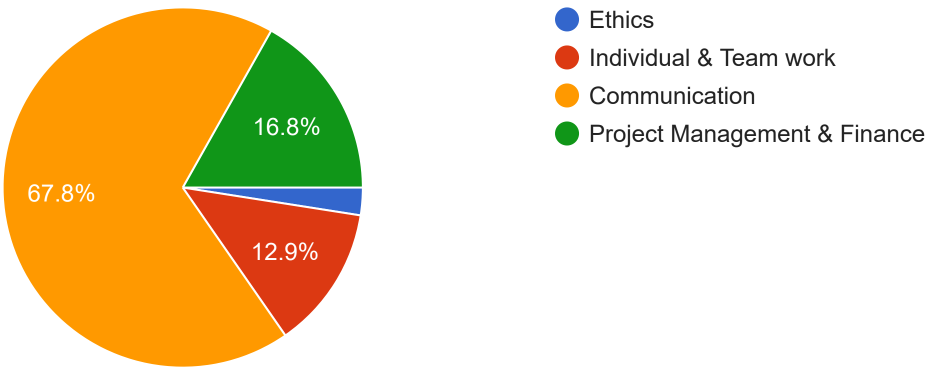
Indirect attainment level of PO & PSO is NOT determined by ___

202 responses



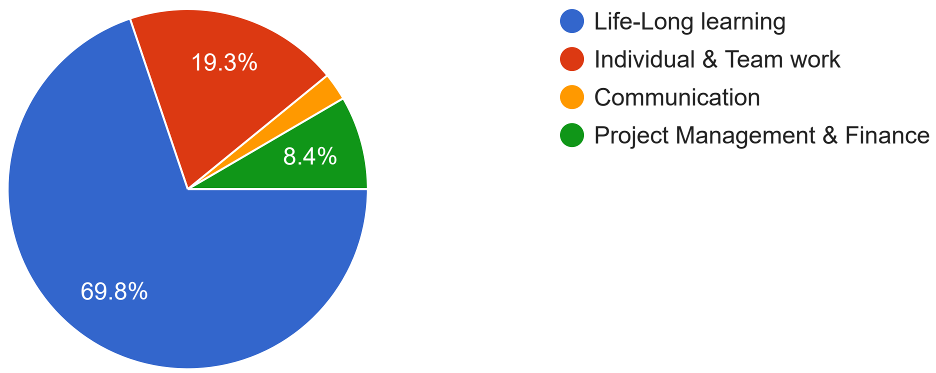
Being able to comprehend & write effective reports & design documentation, make effective presentation will come under PO

202 responses



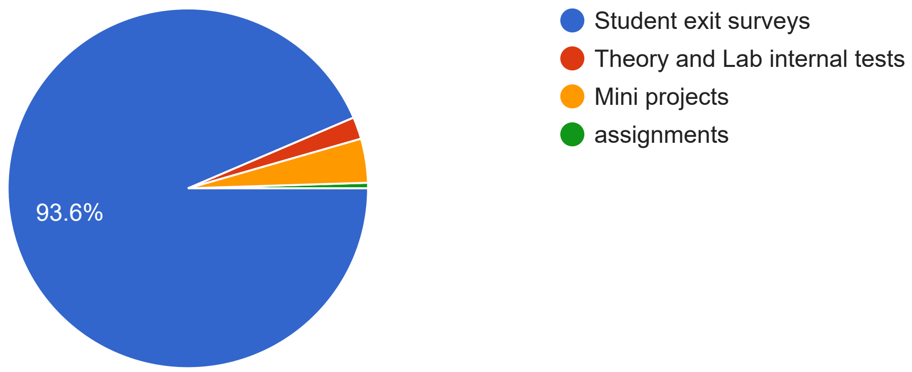
Recognize the need for, and have the preparation & ability to engage in independent in broadest context of technological change will be mapped under Engineering Attributes/PO _____

202 responses



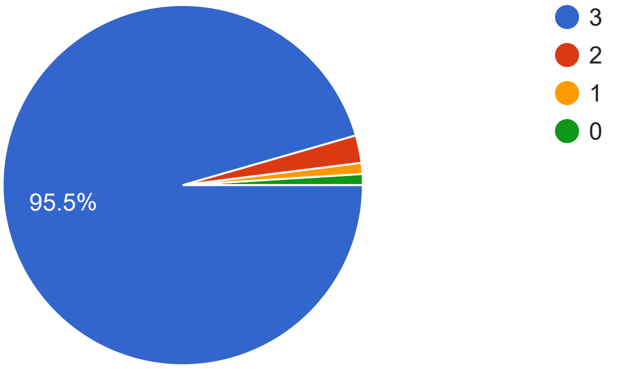
Direct attainment level of PO & PSO is NOT determined by ___

202 responses



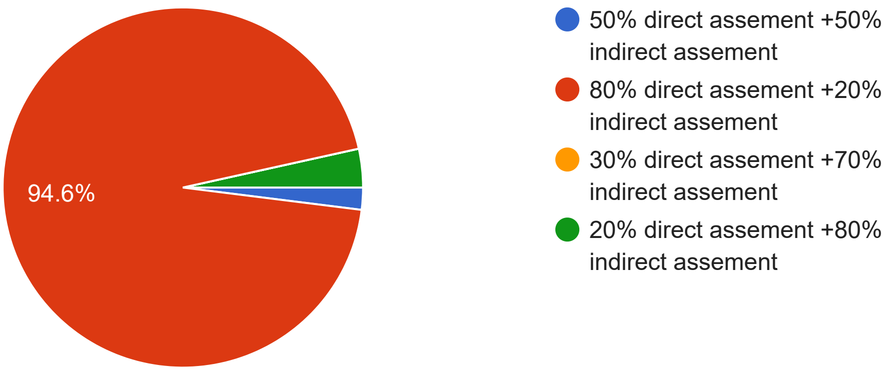
For Course Outcomes attained through result of university examinations >= 80% may be targetted for attainment level ____

202 responses



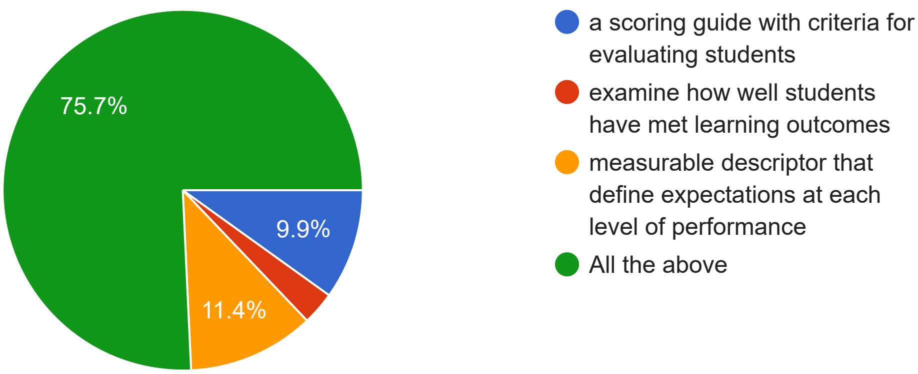
PO attainment level should be calculated by ______

202 responses



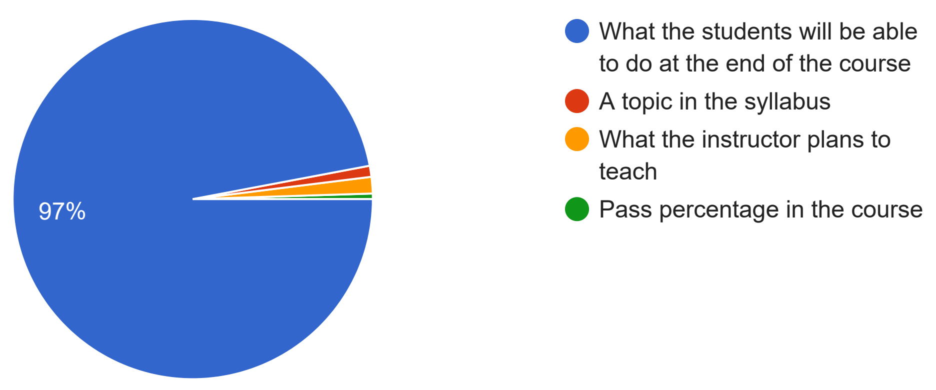
What are Rubrics?

202 responses



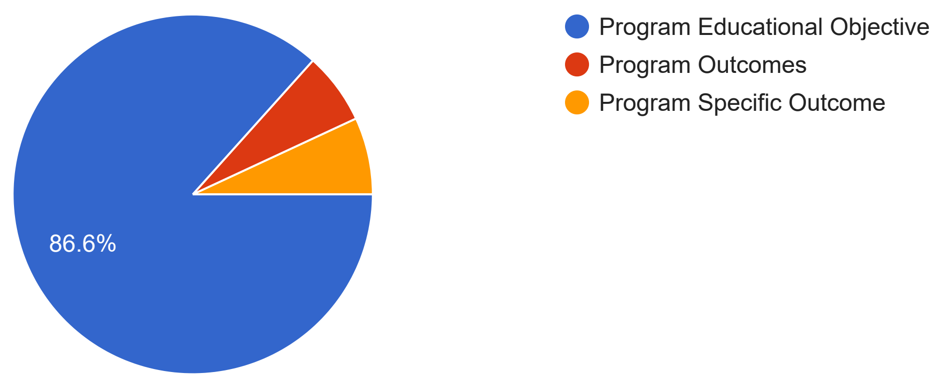
A course outcome is:

202 responses



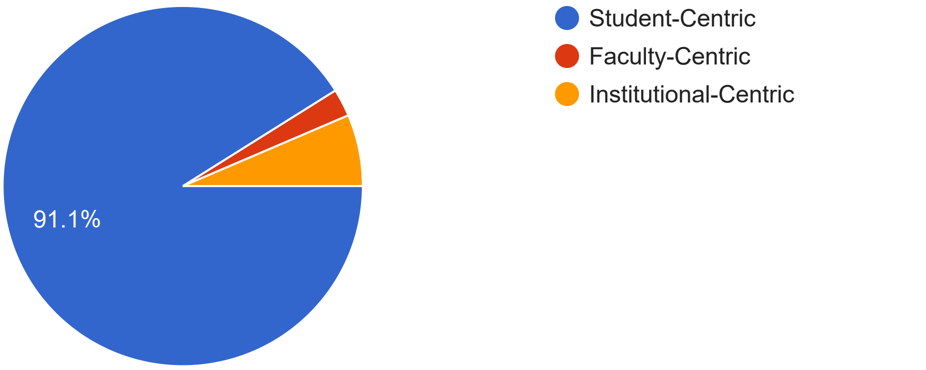
Which of the following are the statements that describe the expected achievements of graduates in their career and also in particular what the graduates are expected to perform and achieve after 4 to 5 years of graduation

202 responses



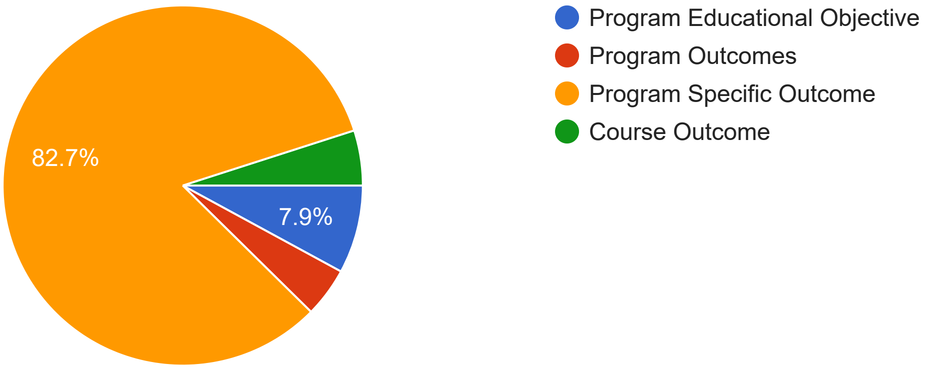
OBE focusses on the below approach

202 responses



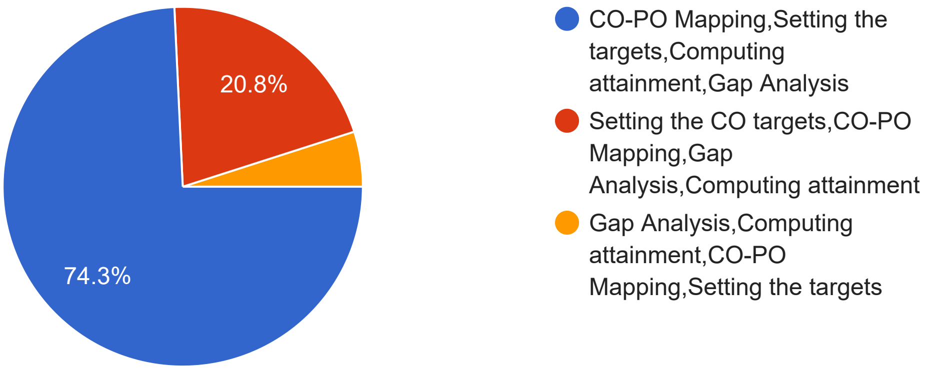
Which one of the below statements are specific to the programme can be 2 to 4 in number

202 responses



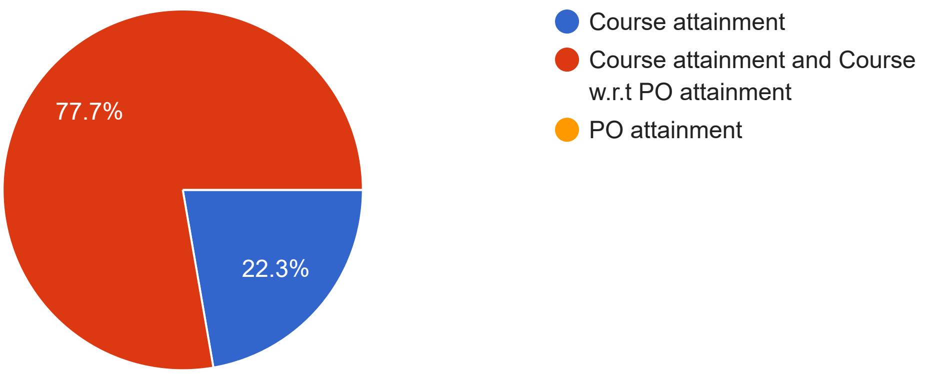
Which of the below statements are correct in CO-PO analysis

202 responses



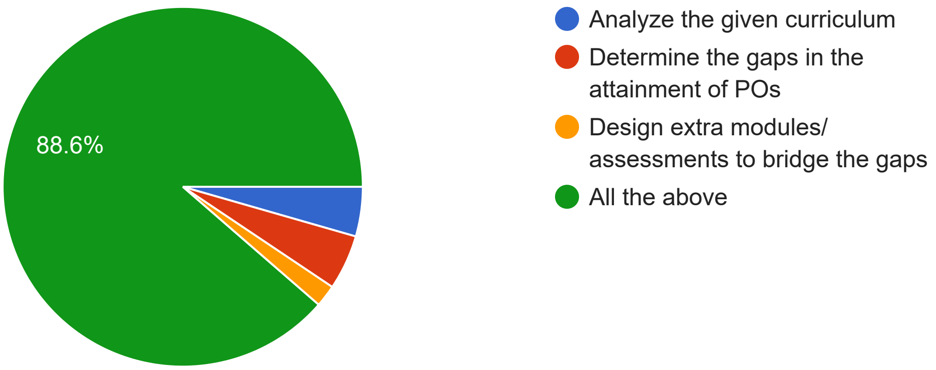
Which of the below attainment needs to be computed by every faculty?

202 responses



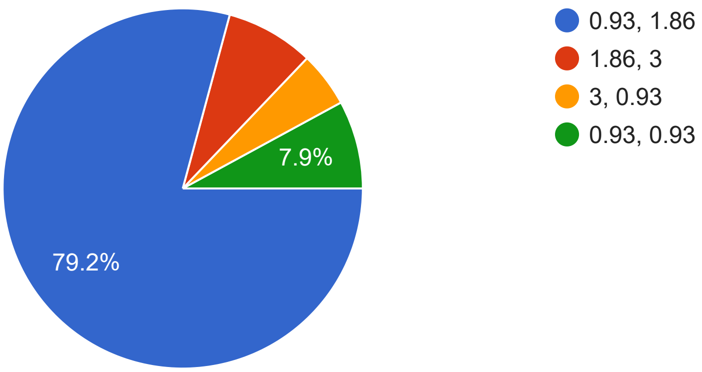
How to attain the POs in view of the given curriculum

202 responses



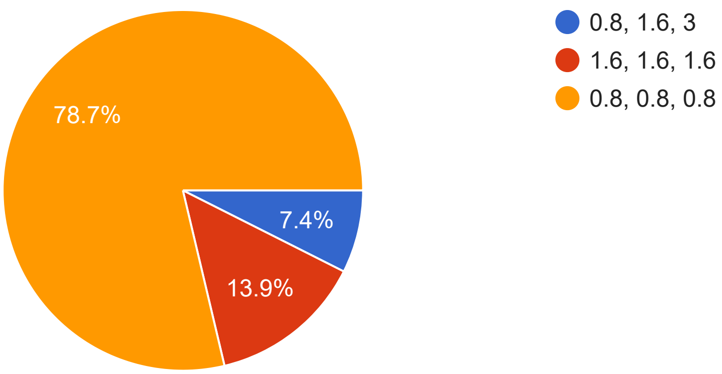
Consider the below CO-PO mapping and the CO attainment values Calculate the attainment of CO1 w.r.t PO1 and PO2

202 responses



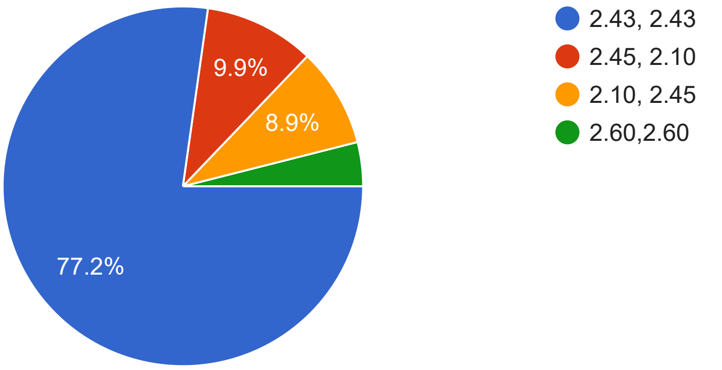
Consider the CO-PO mapping and the CO attainment values as below. Calculate the attainment of CO3 w.r.t PO9,PO10 and PO12

202 responses



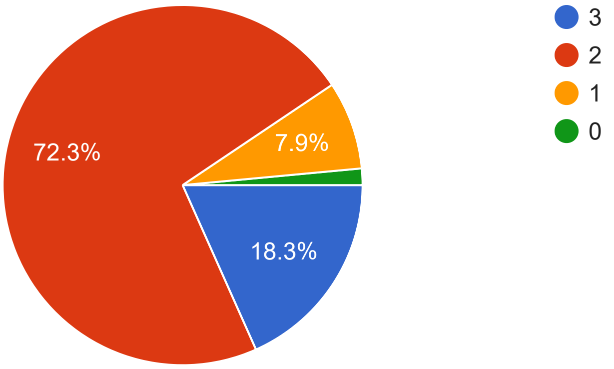
Calculate the overall course attainment of PO10 and PO12 by taking the below values

202 responses



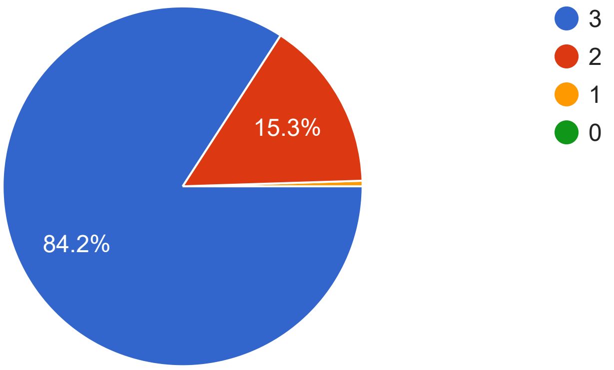
In a class of 50 students if 28 students attempted question1 of max 10 marks and scored 8 marks in IAT1 then the level of attainment in IAT1 is

202 responses



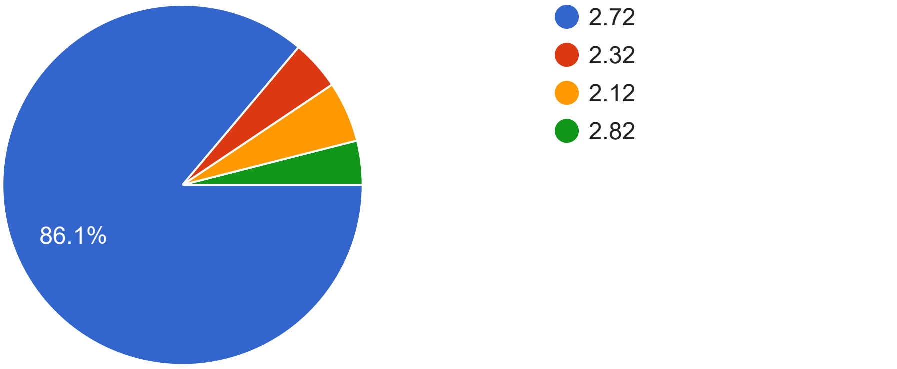
In a class of 103 students if 70 students have scored 60 marks out of 80 in externals then the attainment value is

202 responses



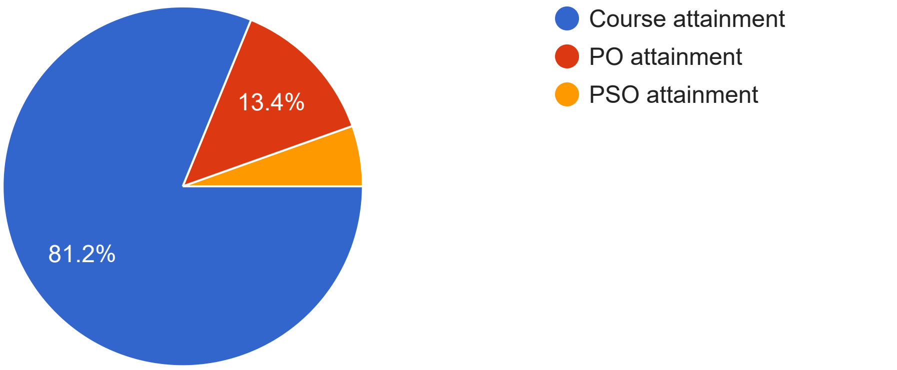
If the internal attainment of a course is 2.6 and external attainment is 2.8 then the overall attainment of that course in 2017 scheme is

202 responses



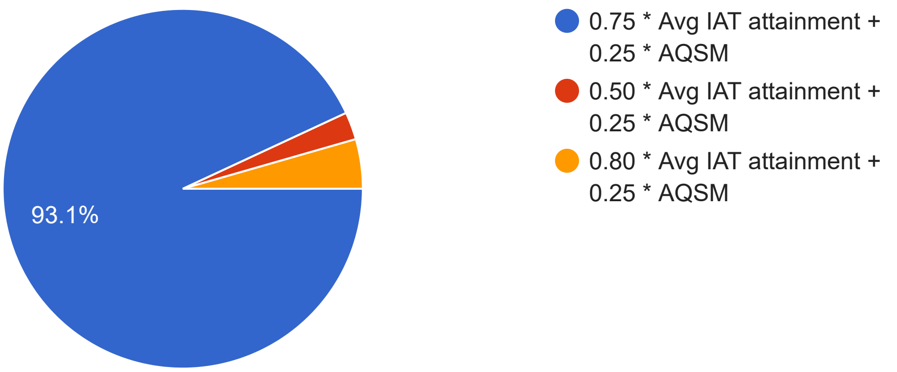
Assignments, mini projects, seminar and Quiz can be used for the computation of the below attainment

202 responses



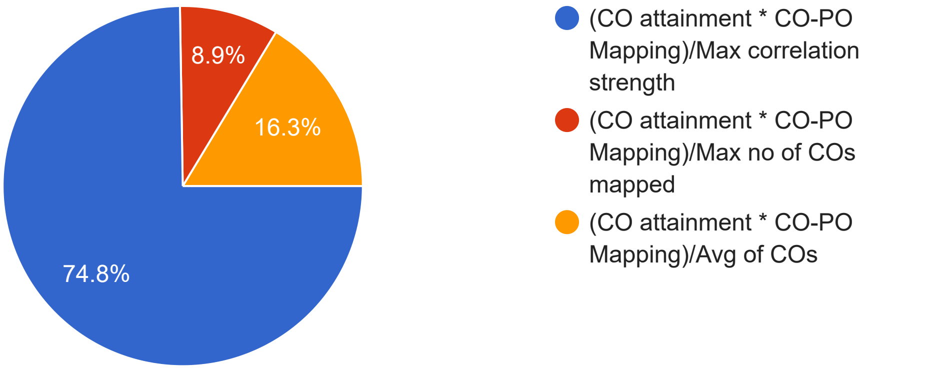
After considering AQSM for 25% the course attainment at internals is

202 responses



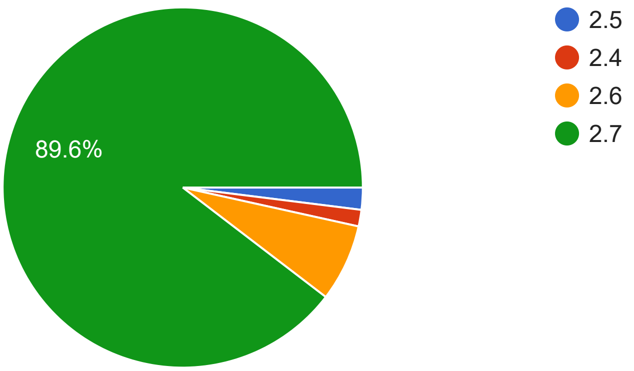
PO attainment can be computed for a batch using the below formula.

202 responses



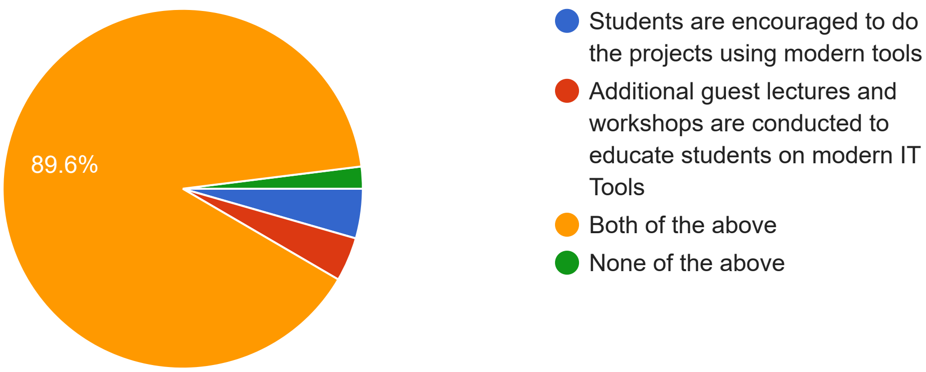
Target set for a course is 2.5 and the course attainment is 2.7.Next time for the same course target can be set to

202 responses



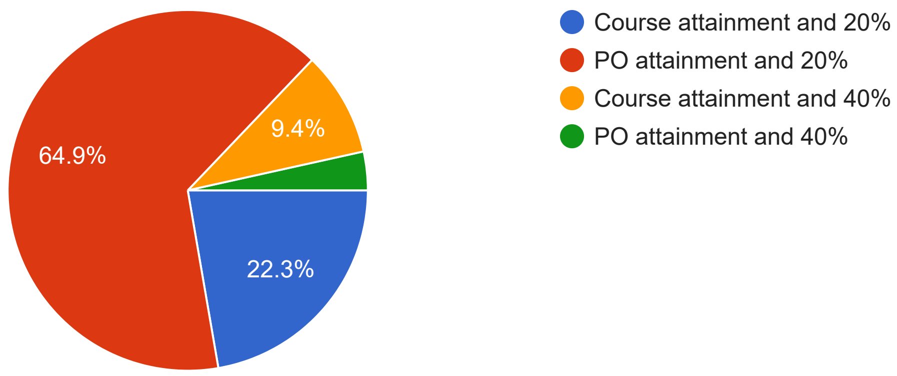
Which of the following action item corresponds to PO5: Modern tool usage after the gap is identified

202 responses



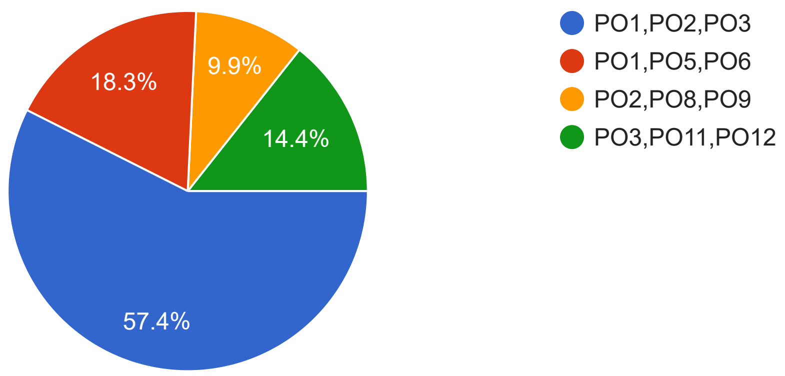
In-direct attainment is used for the calculation of ____ and the weight age given was ------

202 responses



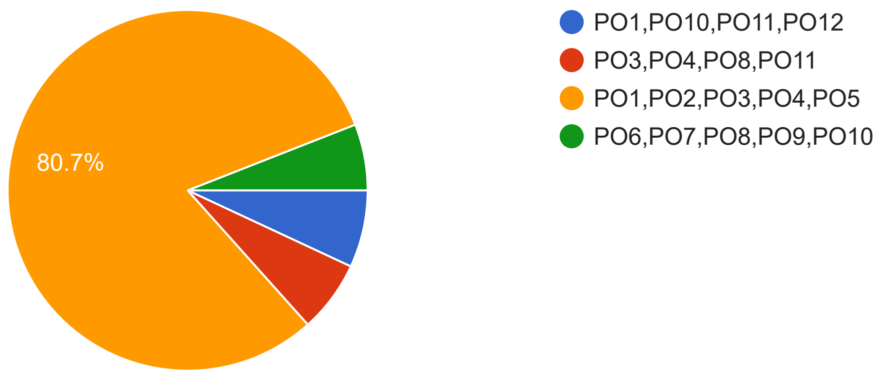
Consider a course outcome as : Describe the behavior of a dynamic system and create an analogous model for a dynamic system. Which of the below POs it is mapped

202 responses



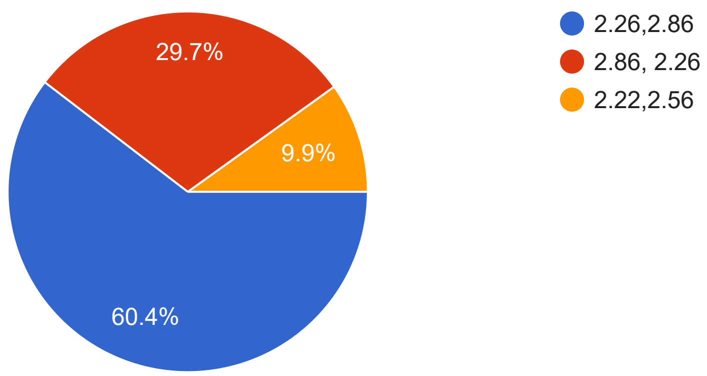
Which of the below POs are considered as domain/program dependent POs

202 responses



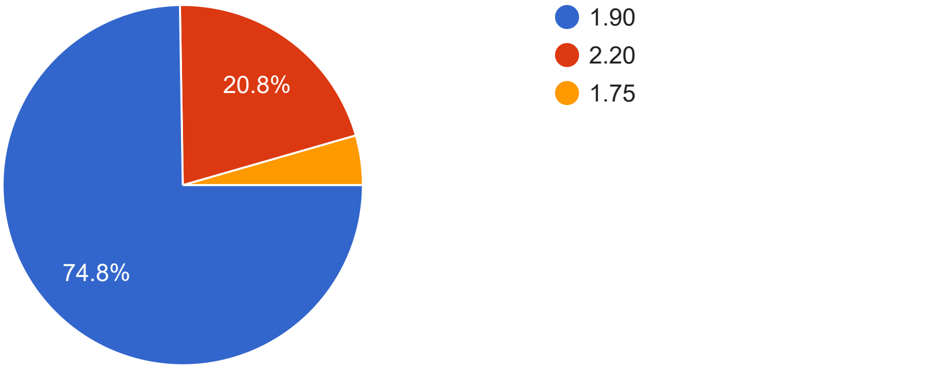
Consider the CO-PO mapping and the CO attainment values as below. Calculate the overall course attainment of PO2 and PO3.

202 responses



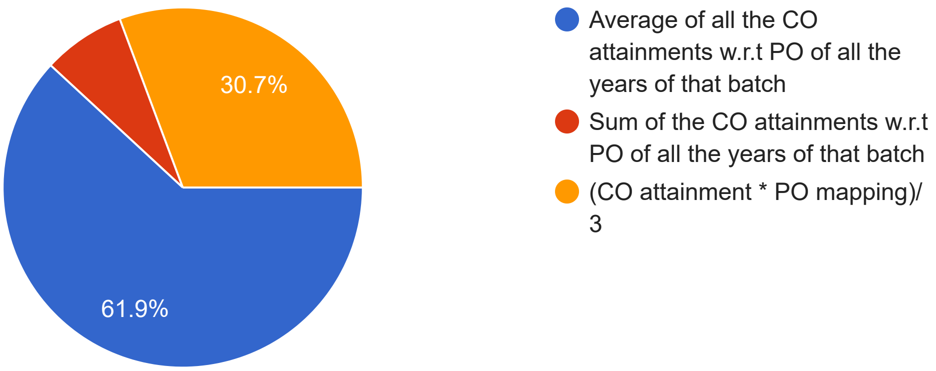
Consider the CO-PO mapping and the CO attainment values as below. Calculate the overall course attainment of PO1.

202 responses



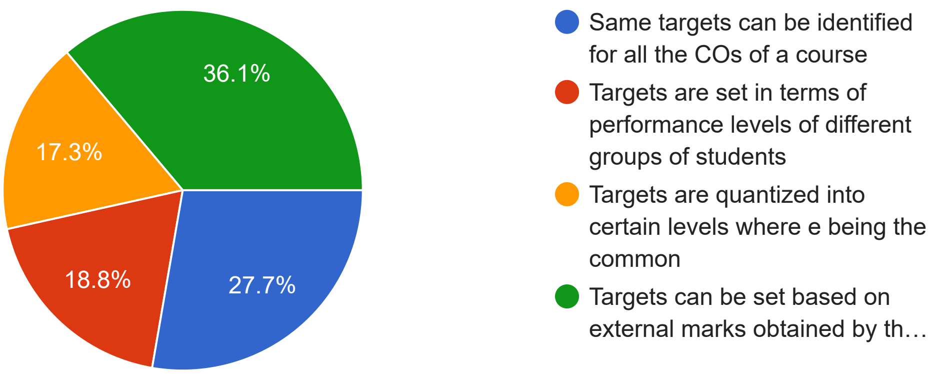
PO attainment of a batch is calculated as

202 responses



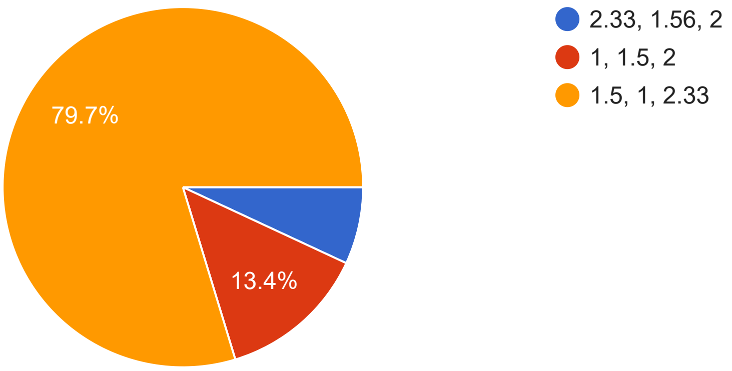
Which one is incorrect among the below statements

202 responses



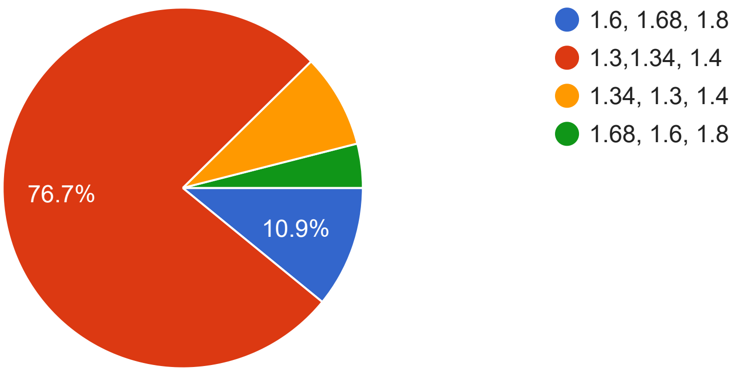
IAT1 questions addresses CO1 and CO3 with an attainment of 2 and 3, IAT2 addresses CO2 and CO3 with an attainment of 1 and 2 and IAT3 addresses CO1 and CO3 with an attainment of 1 and 2. What is the final CO1,CO2 and CO3 attainment?

202 responses



Given CO1 internal and external attainment as 2.5 and 1, CO2 internal and external attainment as 2.7 and 1 and CO3 internal and external attainment as 3 and 1. What is the final CO1,CO2 and CO3 attainment if it is 2015 scheme

202 responses



Mention the MBTI personality type you have got

202 responses

ENTJ

ENTJ - The Commander

ESTJ

ENFJ

ESTJ - The Supervisor

ISTJ - The Inspector

ENFJ - The Giver

ENTJ - The Commander

ENTP - The Visionary